C++ Programming
Course summary | Instructor: Bro Code

A short summary of the course: from your first program to pointers, classes and inheritance.

Week 1-2
Setup and Hello World
Install a compiler (g++, clang or MSVC). Every program starts at main. Include <iostream> and print with std::cout << "Hello" << std::endl;. Compile with g++ main.cpp -o app.

Variables and Data Types
Common types are int, double, char, bool and std::string. Declare with int age = 25;. A variable keeps its type, and const makes a value unchangeable.

Control Flow
Use if / else, switch, for, while and do-while. break leaves a loop and continue skips to the next round. Combine conditions with &&, || and !.

Functions
A function has a return type, a name and parameters. Declare a prototype before use. Parameters are passed by value, or by reference with & to avoid copies. Functions can be overloaded.

Week 3-4
Arrays and Strings
Arrays have a fixed size (int nums[5]) and start at index 0. C++ does not check bounds. std::vector and std::string are safer and more flexible, and offer .size().

Pointers
A pointer stores a memory address. &x gives the address of x and *p reads the value at p. Use nullptr for empty pointers. Prefer smart pointers like std::unique_ptr over raw new and delete.

Classes and Objects
A class bundles data and functions. Constructors initialize objects, and public / private control access. Inside a method, this points to the current object.

Inheritance
A derived class reuses a base class: class Dog : public Animal. virtual functions enable polymorphism, and base classes used polymorphically need a virtual destructor.

Quick reference

std::cout << x	Print a value
std::cin >> x	Read a value
&x	Address of x
*p	Value pointed to by p

nullptr	The empty pointer